Residential Energy Storage Unit

LT5000R

Reference Manual



Our Energy Works For You

LiFeTECH

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1. Safety Precautions

It is very important and necessary to read the user manual carefully (in the accessories) before installing or using battery. Failure to do so or to follow any of the instructions or warnings in this document can result in electrical shock, serious injury, or death, or can damage battery, potentially rendering it inoperable.



Observe these instructions and keep them located near the Li-ion Battery forfuture reference.It is very important and necessary to



Work on a Li-ion Battery should be carried out by qualified personnel only.

1.1. General warnings



While working on the Li-ion Battery wear protective eyeglasses and clothing.



Any uncovered battery material such as electrolyte or powder on the skin or in the eyes must be flushed with plenty of clean water immediately. Then seek medical assistance. Spillages on clothing should be rinsed out with water.



Explosion and fire hazard. Terminals of the Li-ion Battery are always alive; therefore, do not place items or tools on the Li-ion Battery. Avoid short circuits, too deep discharges and too high charge currents. Use insulated tools. Do not wear any metallic items such as watches, bracelets, etc. In case of fire; you must use a type D, foam or CO2 fire extinguisher.



Do not open or dismantle the battery. Electrolyte is very corrosive. In normal working conditions contact with the electrolyte is impossible. If the battery casing is damaged do not touch the exposed electrolyte or powder because it is corrosive.



Li-ion batteries are heavy. If involved in an accident they can become a projectile! Ensure adequate and secure mounting and always use suitable handling equipment for transportation.



Handle with care because a li-ion battery is sensitive to mechanical shock.



Do not expose cable outside, all the battery terminals must be disconnected



Do not place at a children or pet touchable area.



Do not use cleaning solvents to clean battery.



Do not expose battery to flammable or harsh chemicals or vapors.


Do not paint any part of battery; include any internal or external components.


Do not drop, deform, impact, cut or spearing with a sharp object.


Do not wet the battery and avoid of direct sunlight.


Do not use a damaged battery.


Please contact the supplier within 24 hours if there is something abnormal.


Any foreign object is prohibited to insert into any part of battery.


The warranty claims are excluded for direct or indirect damage due to items above.

1.2. Charge and discharge warnings


If the battery is stored for long time, it is required to charge them every six months, and the SOC should be no less than 90%.


Battery needs to be recharged within 12 hours, after fully discharged.


Do not connect battery with PV solar wiring directly.


Use only with a approved BMS.


If charged after the Lithium Battery was discharged below the “Discharge cut-off voltage”, or when the Lithium Battery is damaged or overcharged, the Lithium Battery can release a harmful mixture of gasses such as phosphate.


The temperature range over which the battery can be charged is 0°C to 55°C. Charging the battery at temperatures outside this range may cause severe damage to the battery or reduce battery life expectancy.


The temperature range over which the battery can be discharged is -20°C to 55°C. Discharging the battery at temperatures outside this range may cause severe damage to the battery or reduce battery life expectancy.

1.3. Transportation warnings



The battery must be transported in its original or equivalent package and in an upright position. If the battery is in its package, use soft slings to avoid damage.



Do not stand below a battery when it is hoisted.



Never lift the battery at the terminals or the BMS communication cables, only lift the battery at the handles.

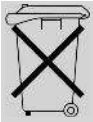
NOTE：

- *Batteries are tested according to UN Handbook of Tests and Criteria, part III, sub section 38.3 (ST/SG/AC.10/11/Rev.5).*
- *For transport the batteries belong to the category UN3480, Class 9, Packaging Group II and have to be transported according to this regulation. This means that for land and sea transport (ADR, RID & IMDG) they have to be packed according to packaging instruction P903 and for air transport (IATA) according to packaging instruction P965. The original packaging complies with these instructions.*

1.4. Disposal of lithium batteries



Batteries marked with the recycling symbol must be processed via a recognized recycling agency. By agreement, they may be returned to the manufacturer.



Batteries must not be mixed with domestic or industrial waste.



Do not throw a battery into fire.

1.5. Before Connecting

◆ After unpacking, please check product and packing list first, if product is damaged or lack of parts, please contact with the local retailer;

◆ Before installation, be sure to cut off the grid power and make sure the battery is in the turned-off mode;

◆ Wiring must be correct, do not mistake the positive and negative cables, and ensure no short circuit with the external device;

◆ It is prohibited to connect the battery and AC power directly;

◆ The embedded BMS in the battery is designed for 48V DC, please DO NOT connect battery in series;

◆ Battery system must be well grounded and the resistance must be less than 1oumu;

◆ Make sure the grounding connection set correctly before operation.

◆ Please ensured the electrical parameters of battery system are compatible to related equipment;

◆ Keep the battery away from water and fire.

1.6. In Using

◆ If the battery system needs to be moved or repaired, the power must be cut off and the battery is completely shutdown;

◆ It is prohibited to connect the battery with different type of battery.

◆ It is prohibited to put the batteries working with faulty or incompatible inverter;

◆ It is prohibited to disassemble the battery (QC tab removed or damaged);

◆ In case of fire, only dry powder fire extinguisher can be used, liquid fire extinguishers are prohibited;

◆ Please do not open, repair or disassemble the battery except staffs from or authorized by manufacturer. We do not undertake any consequences or related responsibility which because of violation of safety operation or violating of design, production and equipment safety standards.

2. Introduction

LT5000R series lithium iron phosphate battery is a new energy storage products. LT5000R it can be used to support reliable power for various types of equipment and systems. LT5000R series is especially suitable for application scene of high power,limited installation space, and restricted load-bearing and long cycle life.

LT5000R series has built-in BMS battery management system, which can manage and monitor cells information including voltage, current and temperature. What's more, BMS can help extending cycle life by balancing cells during charging and discharging.

Multiple batteries are allowed to be connected in parallel to expand capacity and power to meet the requirements of longer power supporting duration and higher power consumption.

2.1. Lithium iron phosphate battery

The lithium iron phosphate battery (LiFePO4 or LFP) is the safest of the mainstream lithium battery types. A single LFP cell has a nominal voltage of 3.2V. A 51.2V LFP battery consists of 16 cells connected in series.

LFP is the chemistry of choice for very demanding applications. Some of its features are:

◆ Rugged - It can operate in deficit mode during long periods of time.

◆ High round trip efficiency.

◆ High energy density - More capacity with less weight and volume.

◆ High charge and discharge currents - Fast charge and discharges are possible.

◆ Flexible charge voltages.

The lithium iron phosphate battery is therefore the chemistry of choice for a range of very demanding applications.

2.2. Features

◆ The whole module is non-toxic, non-polluting and environmentally friendly;

◆ Cathode material is made from LiFePO4 with safety performance and long cycle life;

◆ Battery management system (BMS) has protection functions including over-discharge, over-charge, and over-current and high/low temperature;

◆ The system can automatically manage charge and discharge state and balance current and voltage of each cell;

◆ Adopted self-cooling mode rapidly reduced system entire noise;

◆ The module has less self-discharge, up to 6 months without charging it on shelf, no memory effect, excellent performance of shallow charge and discharge;

◆ Working temperature range is from -20 to 55 ℃, (Charging 0~55 ℃; discharging -20~55℃) with excellent discharge performance and cycle life;

◆ A vertical energy storage system. It gets rid of the traditional cabinet or cross installation form. Beautiful design and convenient installation.

◆ Equipped with monitoring LED display, can provide basic data visualization, more convenient for users to observe the operation of the system.

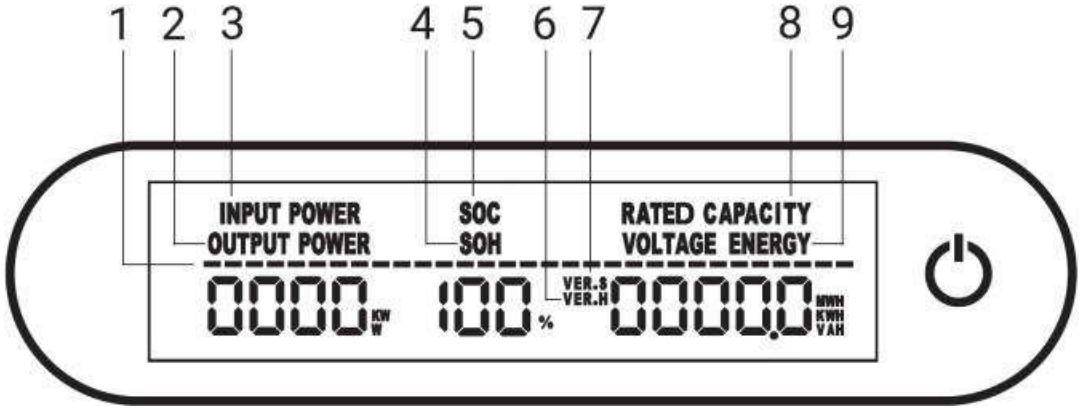
2.3. Specifications



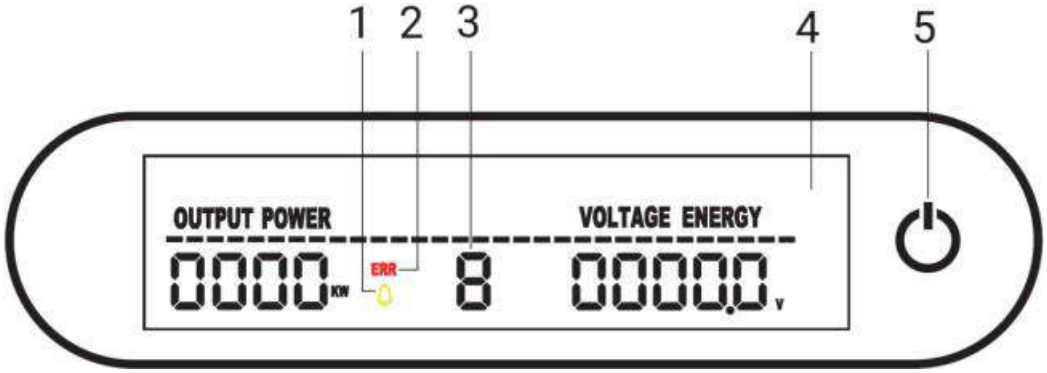
Figure 2.1. Outline dimensional drawing

NO.	Items	Parameters
1	Cell model	100Ah/3.2V
2	Combination Mode	1P16S
3	Nominal Capacity(AH)	100
4	Rated energy(WH)	5120
5	Initial Internal Resistance(m r)	<50
6	Rated Voltage(V)	51.2
7	Charge Cut-off Voltage(V)	59.2
8	Discharge Cut-off Voltage(V)	43.2
9	Standard Charge Current(A)	20
10	Max. Charge Current(A)	≤100
11	Standard Discharge Current(A)	50
12	Max. Discharge Current(A)	≤100
13	Operating Temperature(℃)	-0~+55
		-20~ +55
14	Recommended operating voltage(V)	46.8~56.8
15	Shell type	Painted metal
16	Weight(KG)	50±1
17	Dimension(mm)	600(L)*480(W)*135(H)

2.4. Equipment Interface Instruction

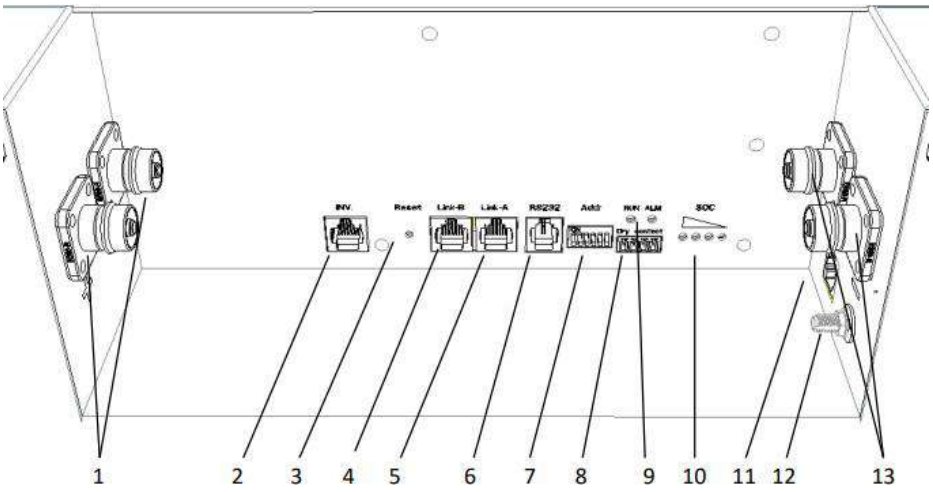


NO.	Instructions	NO.	Instructions
1	Animated streamline	6	Hardware version
2	Discharge power	7	Software version
3	Charging Power	8	Rate capacity
4	SOH	9	Current voltage level or energy throughput
5	SOC		



NO.	Instructions	NO.	Instructions
1	Alarm(warning)	4	Display screen
2	Fault(error	5	Power switch
3	Status Code		

Figure 2.2. Interface definition



NO.	Items	Instructions
1	+	Positive Pole
2	INV. A	CAN/RS485 inverter communication port
3	Reset	Reset button
4	Link-A	Multi-device parallel connection 1
5	Link-B	Multi-device parallel connection 2
6	RS232	RS232 communication port
7	Dry Contact	dry contact
8	Addr	Address Dial Switch
9	RUN&ALM	Run&Alarm indicator
10	SOC	SOC indicator
11	INV. B	CAN/RS485 inverter communication port
12	Wifi/LTE(Opention)	Remote monitoring port
13	-	Remote monitoring port

Power switch

Power switch: press the switch for 3S until the display lights up. When power off, press the switch 5S until the display goes out, BMS standby, no power output.

Display screen

Display screen: the interface can observe the operation status information SOC, SOH, charging and discharging power, alarm fault indication, charging and discharging status display and system status indication of the whole system.

Status code

Status code:When the system status code is displayed as protection information, only the value will be displayed. When the system status code is displayed as fault information, err and fault value will be displayed. The definition of alarm is shown in the table below：

numerical value	Alarm indication	numerical value	Alarm indication
000	No breakdown	009	Low discharge temperature
001	Over voltage protection	011	High ambient temperature
002	Low voltage protection	012	Excessive differential pressure
003	Charging over current protection	013	Discharge circuit failure
004	Discharge over current protection	014	Charging circuit fault
005	Short circuit protection	015	Cell failure
006	Charging high temperature	016	NTC failure
007	High discharge temperature	017	Voltage acquisition fault
008	Low charging temperature	021	Parallel failed

NOTE:

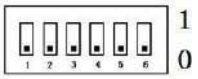
- When the system is charged, the display streamline gathers in the middle, and when it is discharged, the display streamline disperses to both sides

Dry contact

Dry contact: provided 2 ways output dry contact signal.

Address dial switch

ADD Switch: 6 ADD switches, “0”and “1”, refer to picture right.The settings will be active only after restart the battery.



When the battery communicates with the inverter, the address of the battery pack must be set to 1, and the address of the parallel slave should be greater than 1.

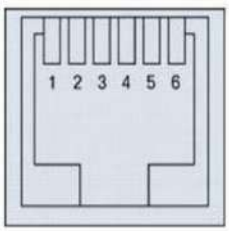
When the battery Pack is connected in parallel, cascading communication is required. Hardware address configuration is required for both the master PACK and the slave PACK, and the hardware address can be set by the dial switch on the board. The definition of the switch refers to the table below.

Address Coding	Dial Code Switch Position						Definition
	#1	#2	#3	#4	#5	#6	
1	ON	OFF	OFF	OFF	OFF	OFF	Set the master Pack, and the inverter communicates with the battery at that address
2	OFF	ON	OFF	OFF	OFF	OFF	Set to the slave Pack1
3	ON	ON	OFF	OFF	OFF	OFF	Set to the slave Pack 2
4	OFF	OFF	ON	OFF	OFF	OFF	Set to the slave Pack 3

5	ON	OFF	ON	OFF	OFF	OFF	Set to the slave Pack 4
6	OFF	ON	ON	OFF	OFF	OFF	Set to the slave Pack 5
7	ON	ON	ON	OFF	OFF	OFF	Set to the slave Pack 6
8	OFF	OFF	OFF	ON	OFF	OFF	Set to the slave Pack 7
9	ON	OFF	OFF	ON	OFF	OFF	Set to the slave Pack 8
10	OFF	ON	OFF	ON	OFF	OFF	Set to the slave Pack 9
11	ON	ON	OFF	ON	OFF	OFF	Set to the slave Pack10
12	OFF	OFF	ON	ON	OFF	OFF	Set to the slave Pack 11
13	ON	OFF	ON	ON	OFF	OFF	Set to the slave Pack 12
14	OFF	ON	ON	ON	OFF	OFF	Set to the slave Pack13
15	ON	ON	ON	ON	OFF	OFF	Set to the slave Pack 14
16	OFF	OFF	OFF	OFF	ON	OFF	Set to the slave Pack 15
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
62	OFF	ON	ON	ON	ON	ON	Set to the slave Pack 61
63	ON	ON	ON	ON	ON	ON	Set to the slave Pack 62

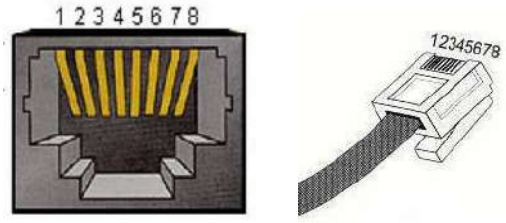
RS232 communication port

RS232 communication port: (RJ11 port) comply with RS232 protocol (baud rate: 9600), for manufacturers or professional engineers debugging or service.

Port definitions	RJ11 Pin	Function
 	1	NC
	2	RS232-GND
	3	RS232-TX
	4	RS232-RX
	5	RS232-GND
	6	NC

Multi-device parallel connection 1 and Multi-device parallel connection 2

Multi-device parallel connection: The same RJ45 port, two RJ45 parallel. Comply with RS485 protocol (baud rate: 9600), used for parallel communication between batteries.

Port definitions	RJ45 Pin	Function
	1	RS485-B
	2	RS485-A
	3	RS485-GND
	4	NC(No connect)
	5	NC(No connect)
	6	RS485-GND
	7	RS485-A
	8	RS485-B

Reset button

◆ When Battery is in sleep mode, press the key for 3S and release it, the Battery will be activated, and the LED indicators will light up from left to right, and then showing the SOC of the battery.

◆ When Battery is in the active state, press the key for 3S and release it, the Battery will go into sleep mode, and the LED indicators will light up from right to left and then all the indicators will be off.

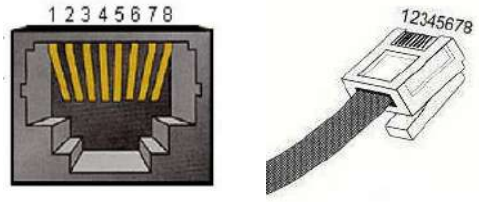
◆ When Battery is in the active state, press the button for 6S and then release, the battery parameters are restored to factory settings, and all LED lights are on for 1.5 seconds.

NOTE:

• *If there are other batteries in the output state in parallel application scenario, the current battery cannot be set to sleep through the reset button at this time, because it will be charged and awakened by other batteries with normal output.*

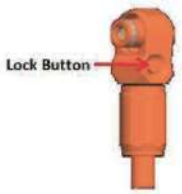
CAN/RS485 communication port

CAN/RS485 communication port: (RJ45 port) follow CAN protocol and RS485 protocol, for output batteries information, the battery uses this interface to communicate with external inverters, PCS and other devices.

Port definitions	RJ45 Pin	Function
	1	RS485-B
	2	RS485-A
	3	NC(No connect)
	4	NC(No connect)
	5	RS485-GND\CAN-GND
	6	NC(No connect)
	7	CANH
	8	CANL

Battery anode and Battery cathode

Battery anode and Battery cathode: there are two pair of terminals with same function, one connect to equipment, the other one paralleling to other battery module for capacity expanding. For each single module, each terminal can achieve charging and discharging function. For power cables uses water-proofed connectors. It must keep pressing this Lock Button while pulling out the power plug.



2.5. Sleep and Wake up

2.5.1 Sleep

When any of the following conditions is met, the battery enters the low-power mode:

1) Under voltage protection is not released within 30 seconds.

2) Press the reset button for 3 seconds and then release the button.

NOTE:

• *If there are other batteries in the output state in parallel application scenario, the current battery cannot be set to sleep through the reset button at this time, because it will be charged and awakened by other batteries with normal output.*

3) The lowest cell voltage is lower than the sleep voltage, and the duration reaches the sleep delay time (while meeting the requirements of no communication, no protection, no equilibrium, and no current).

4) Standby mode lasts for more than 24 hours (no communication, no charge and discharge, no mains power, minimum cell is less than 3.2V).

5) Forced shutdown from the Ems Tools.

Before entering sleep, make sure no charger is connected, otherwise it will not be able to enter low-power mode.

2.5.2Wake up

When the system is in the low-power mode and any of the following conditions is satisfied, the system will exit the low-power mode and enter the normal operation mode:

1) Connect the charger, and the output voltage of the charger must be greater than 48V.

2) Press the reset button for 3 seconds and release the button.

3) Connect the communication line and open the Ems Tools (if enters sleep mode due to over-release protection, and this method cannot wake up the battery).

4) Use the power software switch.

NOTE:

● *After battery over-discharge protection, it enters the low-power mode, wakes up at a regular time every 4 hours, and starts open switch to charging or discharging. If it can be charged, it will exit the sleep mode and enter the normal charging state. If the auto wake up fails to charge for 10 consecutive times, it will no longer auto wake up. When the system is defined as the end of charging, and the recovery voltage is still not reached after 2 days /48h standby time (standby time set value), it is forced to resume charging until the end of recharging.*

2.6. Forced discharge mode

When the battery is in under voltage protection sleep mode and the minimum battery voltage is greater than 2.0V, press and hold the power button, the display will light up and flash until the display stops flashing and releases the power button, and the battery will enter the forced discharge mode for 5 minutes. In the forced discharge mode, if the battery is charged, it will exit the forced discharge mode and switch to the normal mode. If the discharge current exceeds 20A or there is no charging current within 5 minutes, the battery will enter the sleep mode again.

2.7. Automatic parallel

With automatic parallel function; when the slave battery (address > 1) is powered on, the charge and discharge switch is in disconnect state. When the voltage difference between the slave battery and master battery is less than the condition of "the minimum voltage difference between the slave and the master", the master sends the command to the slave. After the slave receives the command from the master, the charge and discharge switch will be connected and the slave is integrated into the master system to complete the parallel operation.

3. How to use the Ems Tools

3.1. Ems Tools connection

1) Connect the RS232 interface of the battery to the computer using the RS232 communication line (this accessory is an optional accessory, need to be purchased separately from the manufacturer).

2) Unzip the package file of the Ems Tools in the same file directory, pay attention to the directory do not store other files.

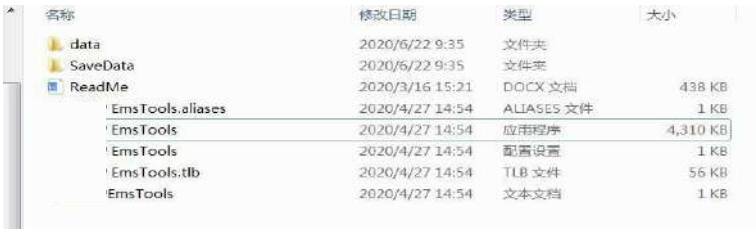


Figure 3.1. Unzip of Ems Tools

3) Open the Ems Tools icon, enter the Protocol selection interface, select the EMS Pack Protocol version and enter the password (please contact the manufacturer for the password) to log in the software.

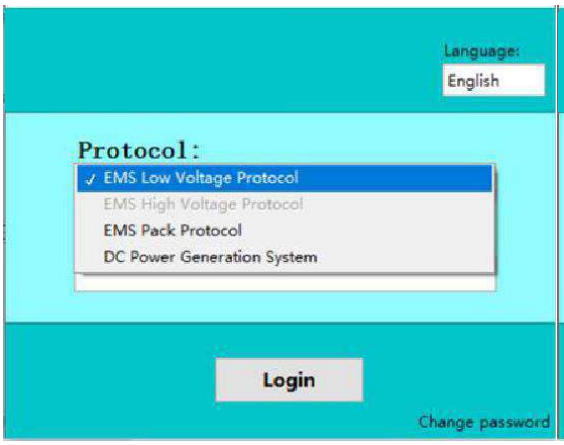


Figure 3.2. Protocol selection interface

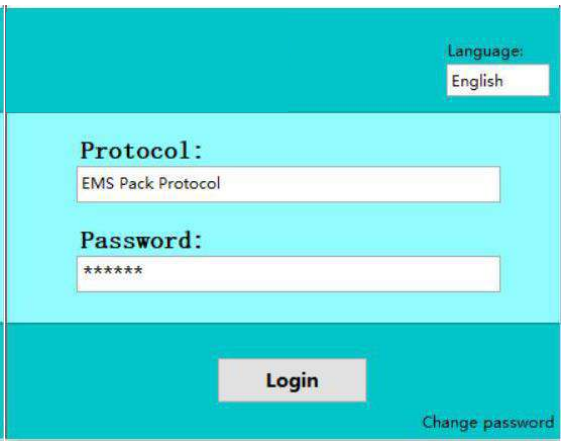


Figure 3.3. Enter the password

4) Users can set different languages according to their own needs. We support four languages, which are Simplified Chinese, English, Japanese and Spanish.



Figure 3.4. Ems Tools language selection

5) Select the serial port number in the EMS low-voltage version of the Ems Tools, and the baud rate is 9600 by default. Click “Open COM” and "Monitor ON" button.

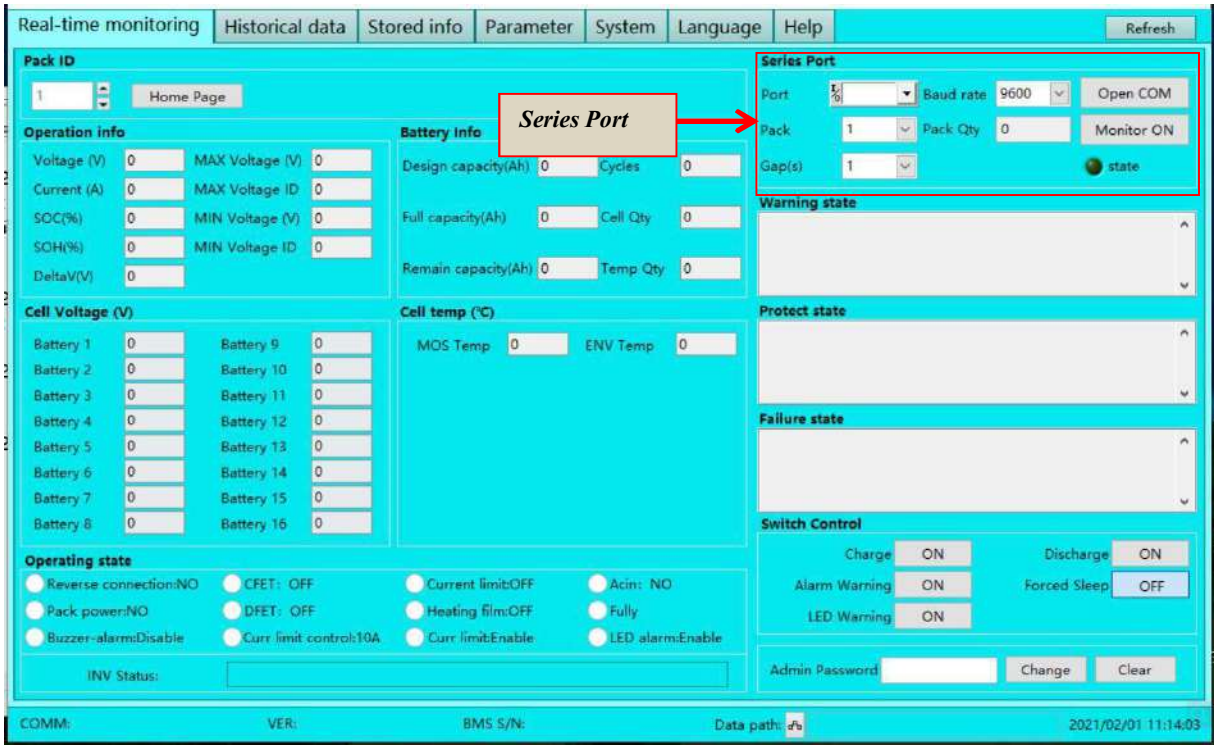


Figure 3.5. Ems Tools serial port settings

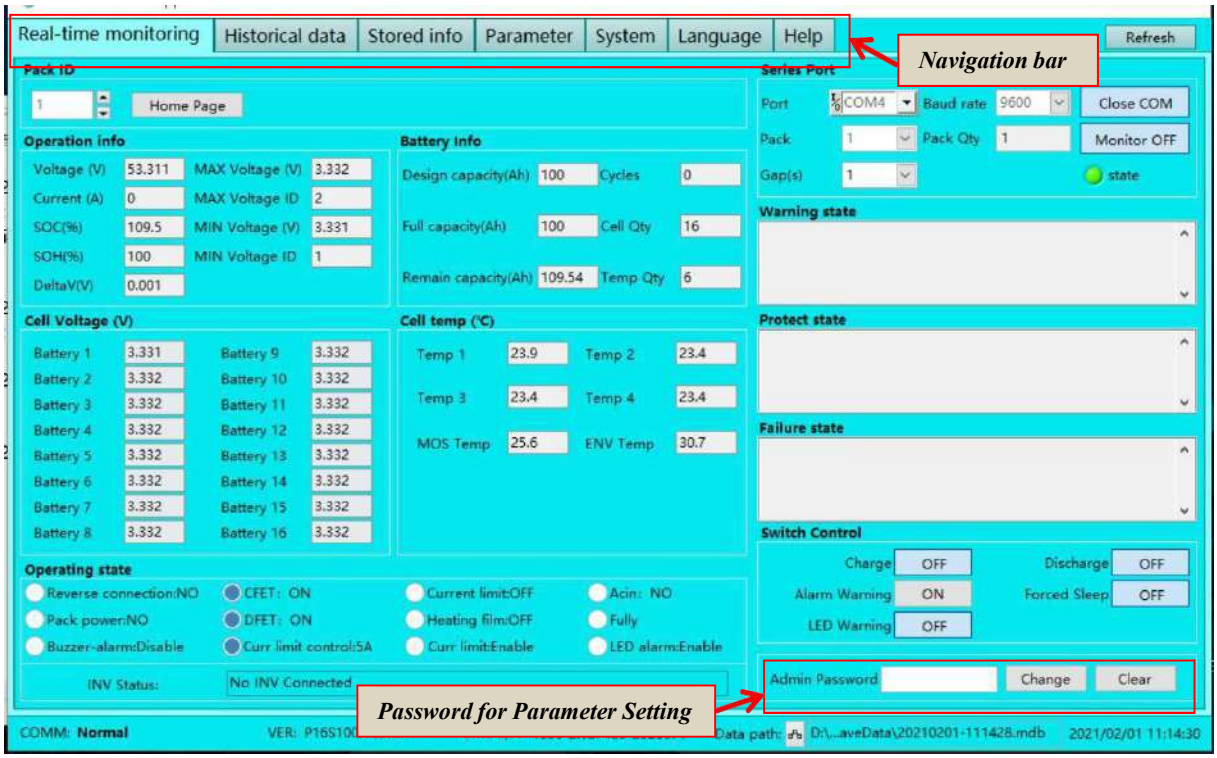


Figure 3.6. Ems Tools data acquisition

6) The corresponding functions can be selected through the navigation bar of the EMS Pack.

7) Historical data recording

Ems Tools supports real-time data recording function, which can record the parameters and state information of the battery when it is running.

By clicking the Historical data menu to switch, all the historical data which collected by Ems Tools can be read at any time, those historical also can be export to computer and save into Excel format files.

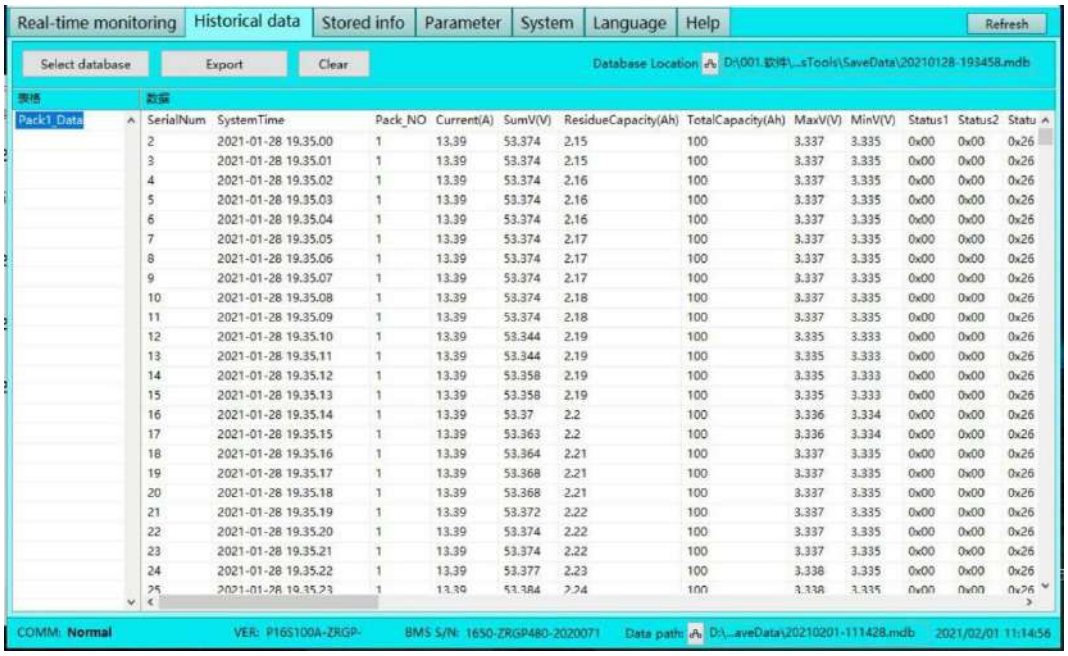


Figure 3.7. Ems Tools historical data recording

8) Log recording

The BMS inside the battery pack uses FLASH chip to record the battery operation data, these data mainly include some parameters such as alarm, protection, fault, state switch, etc., and after-sales personnel can evaluate and analyze the running state of the product according to this log information, battery can support log recording for more than 10 years at most.



Figure 3.8. Ems Tools Log recording

9) Parameter Setting

EMS Pack supports the setting and modification of battery alarm, protection and other parameters. The user needs to enter the password in the lower right corner of the Real-time Monitoring page (the password needs to be obtained from the manufacturer) to modify and set these parameters, refer to Figures 3.6 and Figures 3.9.

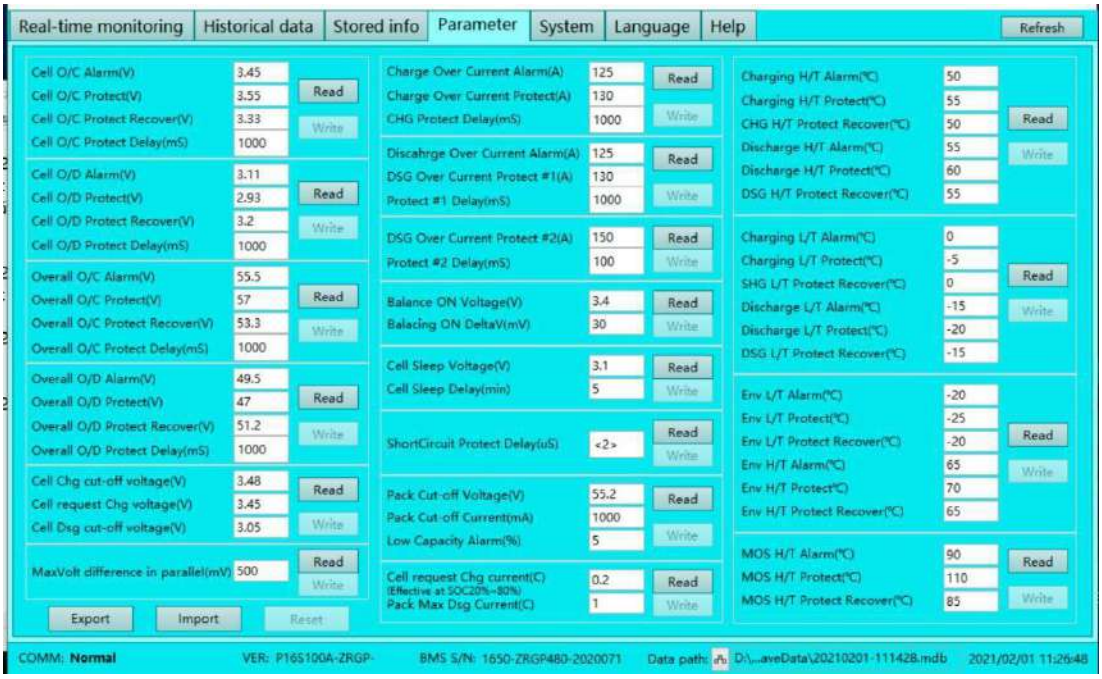


Figure 3.9. Ems Tools Parameters Setting

10) Language Setting

In addition to setting the language on the login interface, users can also set it in the “language” option card in the navigation bar, as shown below:



Figure 3.10. Ems Tools Language Setting

4. How to match communication with inverter

4.1. Supported brands

At present, the energy storage products of our company have completed matching tests with some series inverters of the following brands, and we will continue matching tests with inverters of other companies.



4.2. Inverter matching list

The list tab only lists the inverter manufacturers one of the same series products, general inverter manufacturers in the same series of other products, the communication protocol are the same, so our battery can be communicated with the other products of same series inverter, if encounter a series of products can't communication, please contact us.

The following inverter matching list may not be up to date. The list may change according to the software version upgrade, and the reference manual may does not change immediately according to the software version upgrade. Therefore, if the user wants to get the latest inverter matching support,.

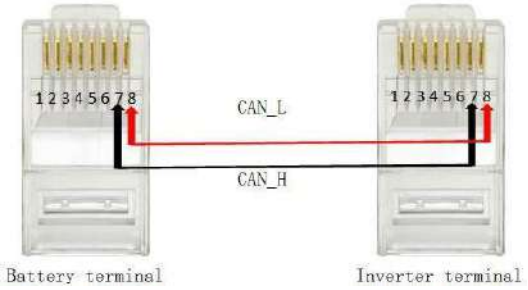
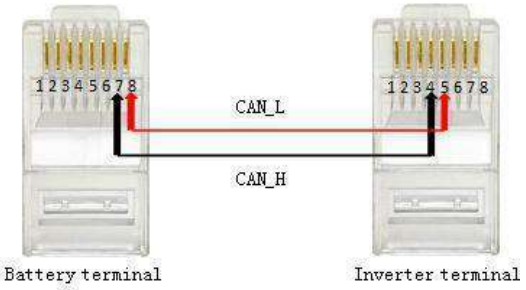
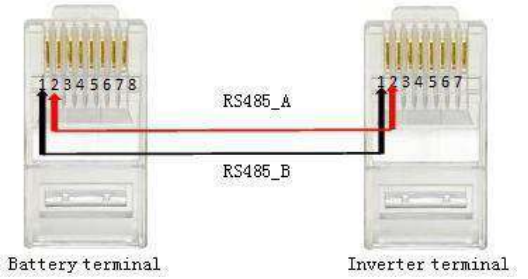
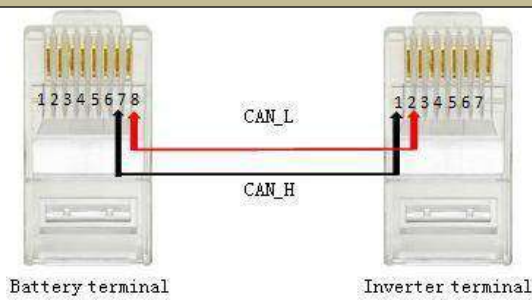
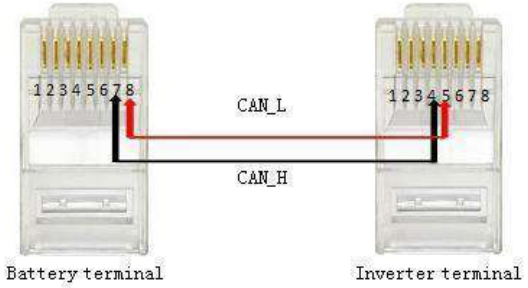
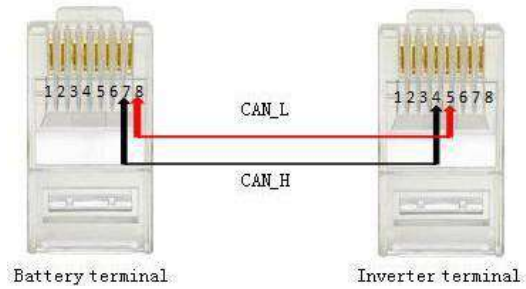
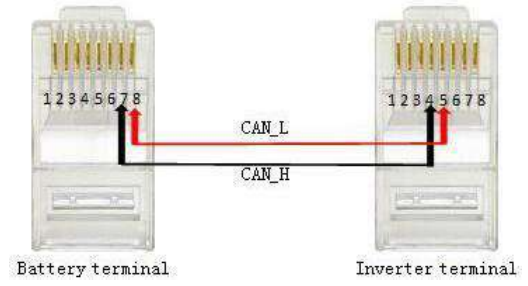
The inverter manufacturer may upgrade its software version, which may cause problems in the communication between our battery and the inverter. Therefore, before communicating with the inverter, please confirm whether the software version of the inverter is consistent with the list. If not, please contact us.

Inverter			LT5000R Firmware version	Communication mode
Brand	Type	Protocol Version		
Growatt	SPF 12KT HVM	1.22	V103	RS485
Studer	Xtender-XTH-8000-48	V1.0.3		Xcom-CAN
Sofar	HYD5000-ES	V6.0		CAN
Solis	RHI-5K-48ES	V1.2		CAN
Goodwe	GW5048-EM	V1.5		CAN
Victron	MultiPlus-II	V6.0		CAN
SMA	S16.0H-12	V2.0		CAN
Sermatec	SMT-5K-TL-UN	V1.2		CAN
Schneider	ConextTM Gateway	V2.0		RS485
PYLON	SUNSYNK-5K-SG01LP1	V1.2		CAN
Li_PLUS	ZRStandard	V1.2		CAN

4.3. Connection with inverter

This section will introduce how to hardware connect LT5000R series products. Inverters manufacturers may upgrade their products, resulting in hardware communication interface changes. If communication is not possible in the application according to the following wiring method, please contact with us.

The CAN/RS485 communication port of LT5000R is connected with the communication interface of inverter.

Connect to the Victron inverter	Connect to the PYLON inverter
	
Connect to the Growatt inverter	**Connect to the Sofar inverter**
	
Connect to the Solis inverter	**Connect to the Goodwe inverter**
	
Connect to the Sermatecl inverter	
	

NOTE:

● *The above CAN and RS485 communication connections are not connected the ground wire, in the application of relatively large interference, it is recommended to connect the ground wire, the ground wire connection method is a single-ended shielding line.*

5. Safe handling of lithium batteries Guide

5.1. Schematic Diagram of Solution

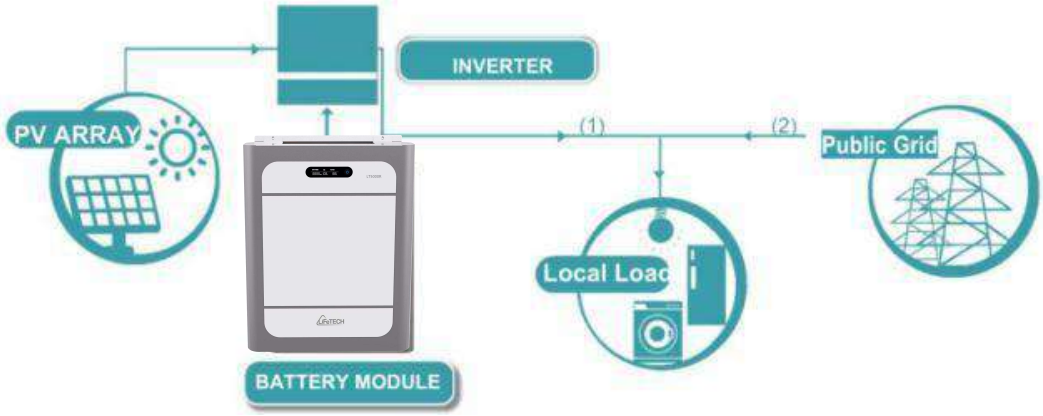


Figure 5.1. Schematic diagram of solution

5.2. Familiar with batteries

Be careful when opening the battery pack. The battery is heavy. Don't lift it with a pole. The weight of the battery can be found in the chapter "specifications".

Familiar with batteries. The battery poles are located on both sides directly behind the battery. The battery polarity is shown on both sides of the battery. The positive pole is represented by "+" and the negative pole by "-".



Figure 5.2. Side view of LT5000R Battery



Figure 5.3. Front view of LT5000RBattery

5.3. Precautions before installation

Before installation, be sure to read the contents in Chapter 1 Safety Precautions, which is related to the operation Safety of installation personnel, please pay attention to.

5.4. Tools

The following tools are required to install the battery pack:



Wire cutter



Cable clamp



Screw Driver

NOTE:

● *Use properly insulated tools to prevent accidental electric shock or short circuits.If insulated tools are not available, cover the entire exposed metal surfaces of the available tools, except their tips, with electrical tape.*

5.5. Safety Gear

It is recommended to wear the following safety gear when dealing with the battery pack:



Insulated gloves



Safety goggles



Safety shoes

6. Installation

6.1. Package Items

Unpacking and check the Packing List:

1) Connector

Each battery will be equipped with a positive connector and a negative connector, the two connectors are not connected to the cable, and the user can be wired according to the actual application needs.



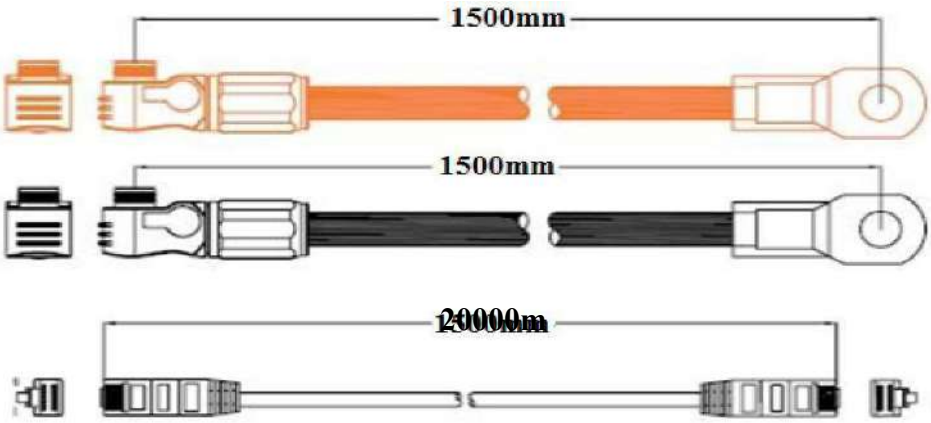
Positive connector



Negative connector

2) Connector For battery module package

Two long power cables and one communication cable for each battery package:



6.2. Installation Location

Make sure that the installation location meets the following conditions:

- ◆ The area is completely water proof.
- ◆ The floor is flat and level.
- ◆ There are no flammable or explosive materials.
- ◆ The ambient temperature is within the range from 0°C to 50°C.
- ◆ The temperature and humidity is maintained at a constant level.
- ◆ There is minimal dust and dirt in the area.
- ◆ The distance from heat source is more. than 2 meters
- ◆ The distance from air outlet. of inverter is more than 0.5 meters.
- ◆ Do not install outside directly
- ◆ Do not cover or wrap the battery case or cabinet.
- ◆ Do not place at a children or pet touchable area.
- ◆ The installation area shall avoid of direct sunlight.
- ◆ There is no mandatory ventilation requirements for battery module, but please avoid of installation in confined area. The aeration shall avoid of high salinity, humidity or temperature.

CAUTION

If the ambient temperature is outside the operating range, the battery pack stops operating to protect, itself. The optimal temperature range for the battery pack to operate is 0°C to 50°C. Frequent exposure, to harsh temperatures may deteriorate the performance and life of the battery pack.

6.3. Wall mounting installation

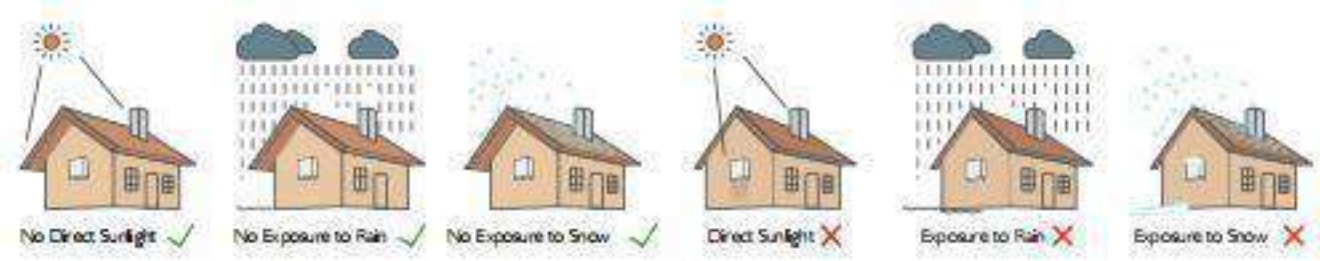
Selection of installation position：

1. The LT5000R shall be mounted on a solid surface suitable for the size and weight of the inverter.

2. The LT5000R shall be installed vertically or on a slope with a maximum of 15 degrees。3..

The ambient temperature should be lower than 45 ℃

4. The installation of LT5000R should avoid direct sunlight, snow, rain, lightning and other bad weather.



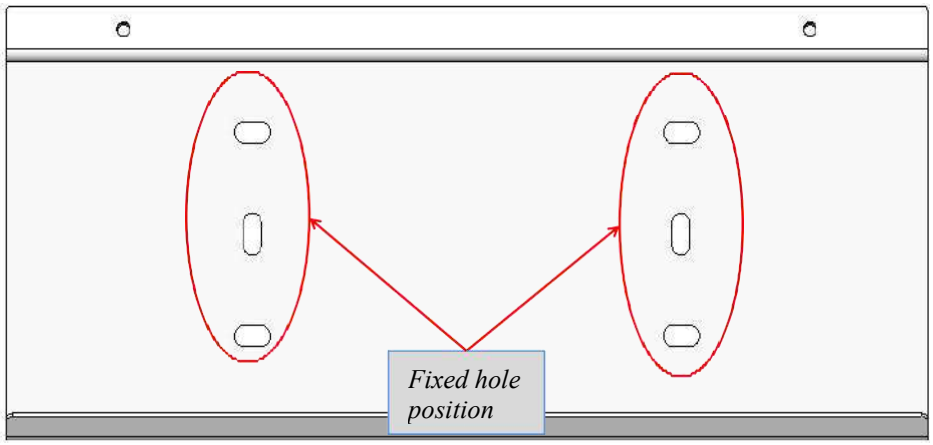
5. LT5000R should be installed at eye level for easy maintenance.

6. The product label on the LT5000R should be clearly visible after installation.

7. Leave enough space around LT5000R as shown in Figure 3.

Install

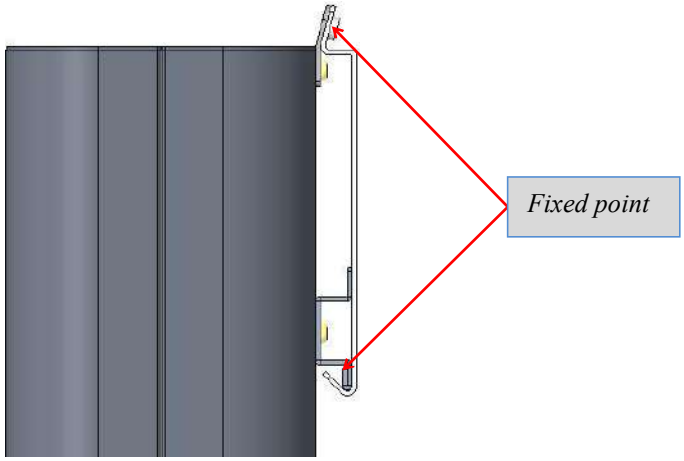
1. Please use the mounting bracket as a template, drill 6 holes (diameter 10 mm, depth 80 mm) in the correct position, and use the expansion bolts in the accessory box to firmly fix the mounting bracket on the wall



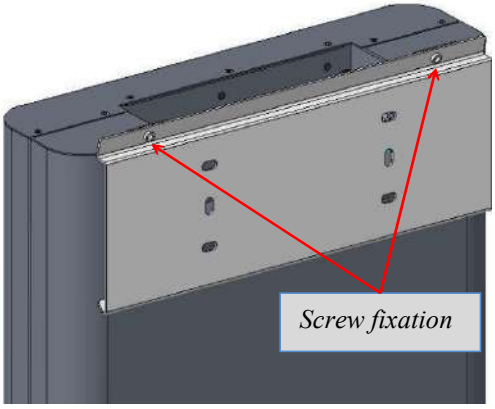
NOTE:

Bearing capacity of the wall must be higher than 17KG, otherwise may not be able to keep LT5000R from dropping.

2. During installation, the system hook shall be firmly attached to the bracket, and the hole position shall correspond to the bracket fixing point。



3. Align the fixing point, install the fixing screw, and fix the screw firmly with the bracket



NOTE:

The system cannot be installed near flammable, explosive or strong electric magnetic field. Remember this system is heavy! Please be careful when removing from the package

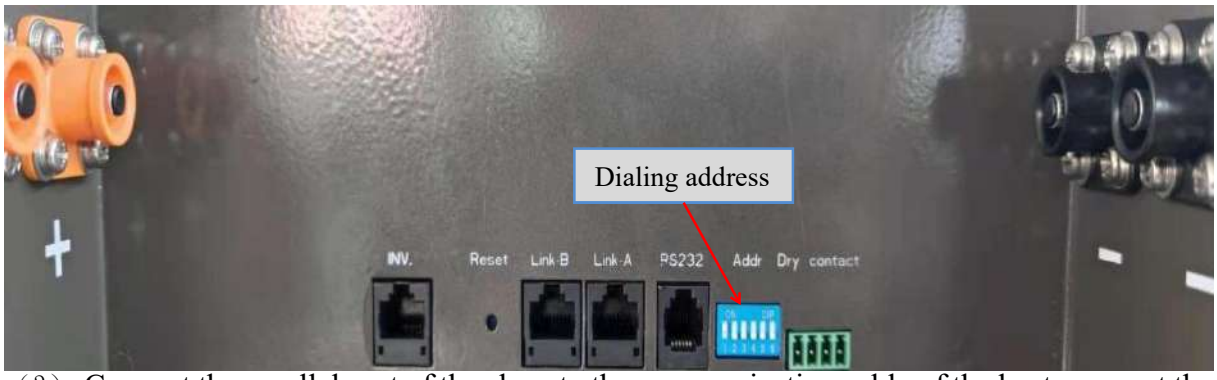
6.4. Parallel Installation

A. Connect the whole system to the cable

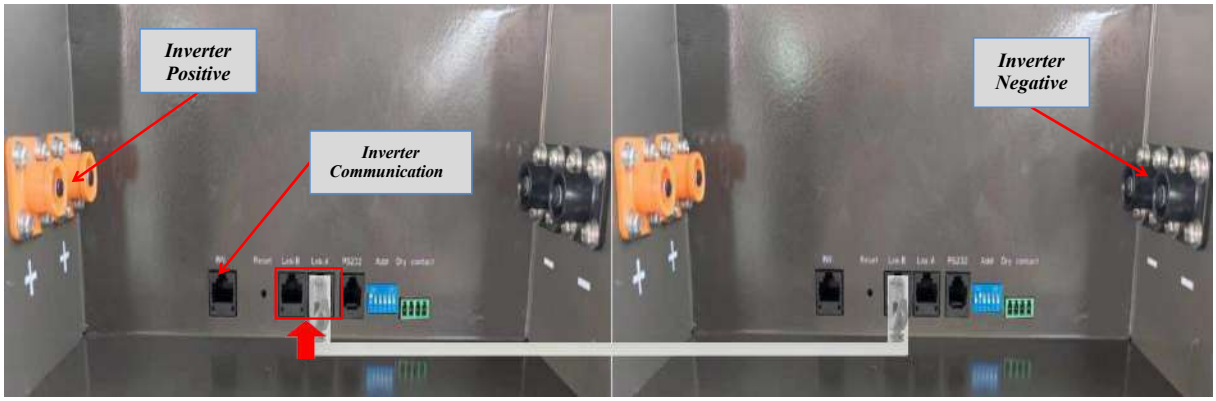
(1) When the system is connected, make sure that the system is shut down. If it is shut down, press and hold the key for 5S until the display is shut down.



（2） The module with the dialing switch of 1 is the master battery module, and the other modules are the slave battery module (one master battery module can be configured with up to 62 slave battery modules). The inverter must communicate with the battery module with the address of



（3） Connect the parallel port of the slave to the communication cable of the host, connect the positive pole of the slave to the positive pole of the host, connect the negative pole of the slave to the negative pole of the host, and finally connect the communication cable of the host to the frequency converter



B. Power On

Double check all the power cable and communication cable.

(1) Press the power switch for a long time, the 3S display will light up and the system will start Switch on all the battery systems:



(2)After power on, the whole system will merge automatically. After merging, all systems will display normally. If the combination fails, the display will show the system disconnection status code 21, which needs to be handled in time.

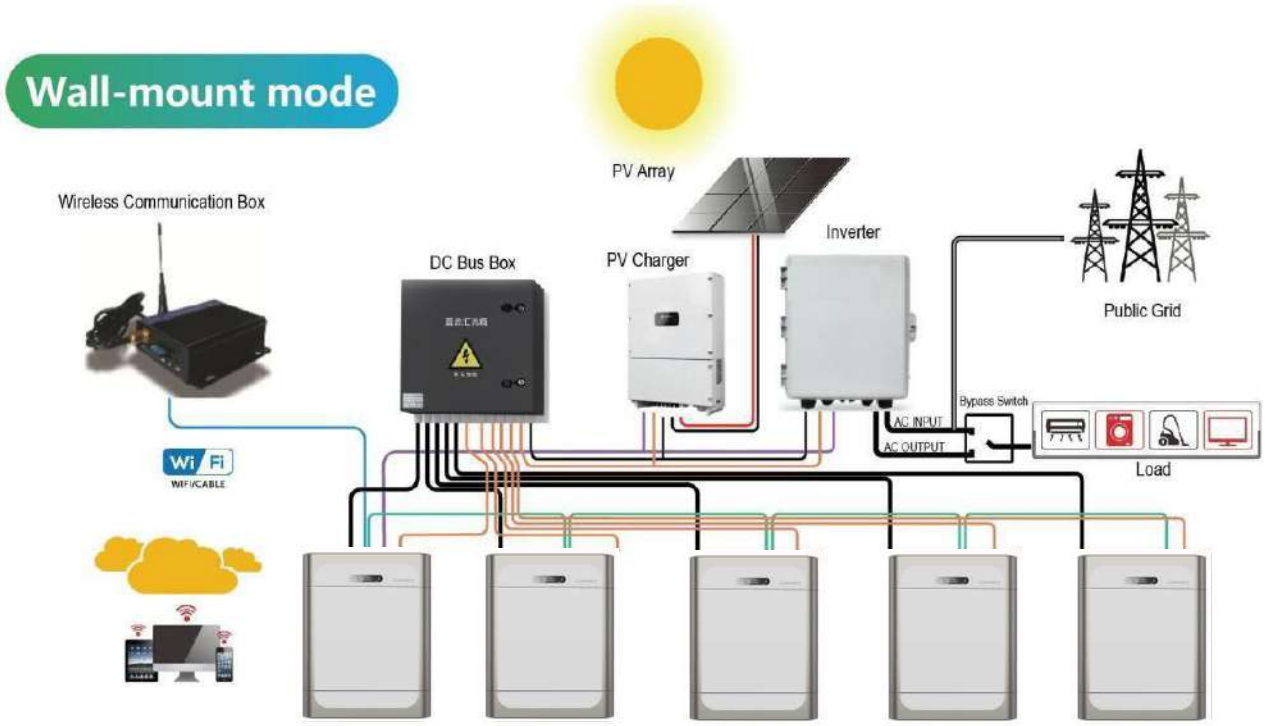


Figure 5.4. Schematic diagram of parallel solution

NOTE:

- To avoid current pulse of the inverter add on the battery bank. Shall start inverter first or switch on breaker between battery and inverter after all connected batteries turned on.
- Between battery bank and inverter should install breaker to protect system safety.
- All the installation and operation must follow local electric standard.

7. Trouble Shooting Steps

7.1. Problem determination based on

1) Whether the battery can be turned on or not;

2) If battery is turned on, check the red light is off, flashing or lighting;

3) If the red light is off, check whether the battery can be charged/discharged or not.

7.2. Preliminary determination steps

1) The system can not be turned on, and the power indicator is not illuminated or flashing. If the external switch of the system is turned on, the running light flashes, and the external power supply voltage exceeds 48V, the system still can not be turned on for operation, please contact the dealer.

2) The system can be turned on, but red light is lighting, and cannot charge or discharge. If the red light is lighting, that means system is abnormal, please check values as following:

a) Temperature: Above 55℃ or under -20 ℃, the system could not work.

Solution: to move system to the normal operating temperature range between 0℃ and 45℃

b) Current: If current is greater than 130A, battery protection will turn on.

Solution: Check whether current is too large or not, if it is, to change the settings on power Supply side.

c) High Voltage: If charging voltage above 57V, battery protection will turn on.

Solution: Check whether voltage is too high or not, if it is, to change the settings on power supply side.

d) Low Voltage: When the battery discharges to 44V or less, battery protection will turn on.

Solution: Charge the battery for some time, the red light turn off

Excluding the four points above, if the faulty is still cannot be located, turn off battery and repair.

7.3. The battery cannot be charged or discharged

1) Cannot be charged:

Disconnect the power cables, measure voltage on power side, if the voltage is 53~54V restart the battery,connect the power cable and try again, if still not work, turn off battery and contact distributor .

2) Unable to discharge:

Disconnect the power cables and measure voltage on battery side, if it is under 44V please charge the battery; if voltage is above 48V and still cannot discharge, turn off battery and contact distributor.

8. Storage,Transportation and Emergency Situations

8.1. Storage

Recharge and maintain the battery pack regularly every three months to ensure the battery is in the best condition.

8.2. Transportation

Battery packs need to be packed before they can be shipped, during transportation, severe impact, extrusion, direct sunlight and rain should be protected.

8.3. Emergency Situations

(1). Leaking Batteries

If the battery pack leaks electrolyte, avoid contact with the leaking liquid or gas. If one is exposed to the leaked substance, immediately perform the actions described below. Inhalation: Evacuate the contaminated area, and seek medical attention.

Contact with eyes: Rinse eyes with flowing water for 15 minutes, and seek medical attention.

Contact with skin: Wash the affected area thoroughly with soap and water, and seek medical attention.

Ingestion: Induce vomiting, and seek medical attention. 2)

(2). Fire

NO WATER! Only dry powder fire extinguisher can be used; if possible, move the battery pack to a safe area before it catches fire.

(3). Wet Batteries

If the battery pack is wet or submerged in water, do not let people access it, and then contact or an authorized dealer for technical support.

(4). Damaged Batteries

Damaged batteries are dangerous and must be handled with the utmost care. They are not fit for use and may pose a danger to people or property. If the battery pack seems to be damaged, pack it in its original container, and then return it to or an authorized dealer.

NOTE:

- Damaged batteries may leak electrolyte or produce flammable gas.
- In case a damaged battery needs recycling, it shall follow the local recycling regulation (ie. Regulation (EC) Nº 1013/2006 among European Union) to process, and using the best availabletechniques to achieve a relevant recycling efficiency